

a Grouping strategies

Exhaustive hypothesis testing

Maree et al. 2024 · $(A!)^C$ per frame



every grouping triangulated + scored

one kept: lowest reprojection error

Greedy per-view assignment

LUC3D · C Hungarian solves, $O(C \cdot A^3)$



each camera assigned once, then committed

targets updated before the next camera

b Association cost

Cost per (target, detection) pair

$\pi(\text{target})$

2D $w_k \cdot \text{corr2d} \cdot (1 - |d - \pi(t)| / \text{velThresh}) \cdot e^{-\lambda \Delta t}$

target t

ray(d)

3D $w_k \cdot \text{corr3d} \cdot (1 - \text{dist}(t, \text{ray}(d)) / \text{distThresh})$

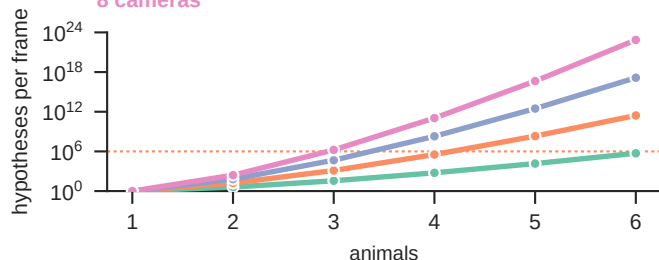
c Hypotheses per frame

2 cameras

4 cameras

6 cameras

8 cameras



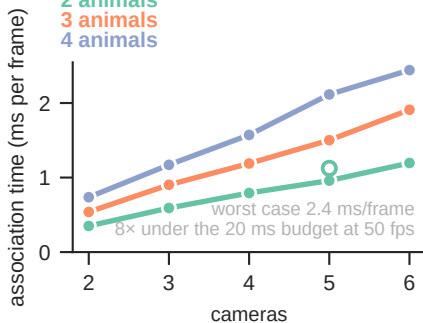
dotted rule: the harness's 10^6 hypotheses/frame cap
greedy (LUC3D) enumerates none: C solves per frame

d Association runtime

2 animals

3 animals

4 animals

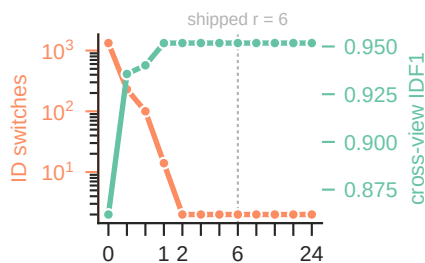


○ second corpus, same rig
one session per line, $n = 1$

e 3D-term ablation

ID switches

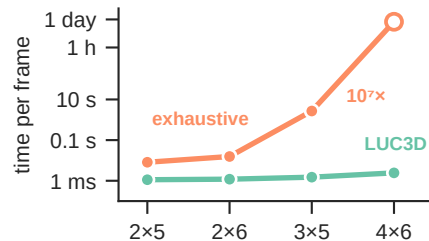
cross-view IDF1



$r = \text{corr3d} / \text{corr2d}$
 $r = 0$: no 3D term at all
both series are LUC3D against itself

f Time per frame

same grouping on 99.999%
of the 137,266 of 198,292
frames exhaustive could run



animals × cameras
○ not run, extrapolated:
 1.9×10^6 hyps × 244–347 μ s

b: the cost function as implemented. c: exact arithmetic. d: measured by scripts/bench/bench_crossview.mjs.

e: 8 BMimica sessions, a fixed 6,000-frame leading window per cell, identical across all 24 cells.

f: exhaustive is our reimplementation of the published per-frame procedure. IDF1 and switches via motmetrics on a shared identity-stripped pool.